

A Comfort Feedback Method for Remote Fitting

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ABSTRACT

Garment e-commerce has become an indispensable part of the garment retail industry. However, the lack of a fitting process in garment e-commerce has led to high return rates and operating costs. The remote fitting method based on the fitting robot helps solve the above problem, but the accuracy of the fitting results needs to be improved to ensure the effectiveness of the remote fitting. The comfort of clothing is the main factor affecting the fitting results. This paper presents a comfort feedback method based on the fitting robot to obtain and feed back the comfort information of remote fitting. This method includes a pressure measuring device, a comfort conversion model, and a real-time feedback interface. And the test results demonstrate that this method is effective. This method's application provides consumers with a new fitting experience and more accurate fitting results, which will promote the development of garment e-commerce.

CCS CONCEPTS

• **Human-centered computing** → **Visualization**; **Human computer interaction (HCI)**; • **Computer systems organization** → **Robotics**.

KEYWORDS

comfort feedback, remote fitting, fitting robot, comfort conversion model

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1 INTRODUCTION

With the rapid development of Internet technology and logistics technology, e-commerce has become a part of people's lives. In the vast e-commerce market, as the garment is a necessity of people's lives, garment e-commerce occupies an important proportion [36].

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Garment e-commerce models include online purchase, online customization, and so on. Due to the lack of the fitting process in garment e-commerce, consumers can only rely on past purchasing experience to choose the size of clothes. However, different manufacturers have different garment production standards, so consumers often buy unsuitable clothes, which leads to an increasingly severe return problem. Statistics show that about 42% of the clothing returns are due to inappropriate sizes [25], which seriously hinders the development of garment e-commerce.

To solve the problem of lacking fitting process in the garment e-commerce, some research teams use the virtual fitting technology [12][26][29] or the fitting robot technology [2][3]. Some researchers have put forward many research results in rapid modeling of human bodies [5][15][33][39], physical simulation of clothing [4][6], fabric simulation [17][38] and so on. However, due to the diversity and complexity of fabrics, there is a certain gap between the simulation effect and the actual effect. Therefore, although some virtual fitting platforms have appeared, they have not been widely used [13]. The fitting robot is a form-changing humanoid robot [3], which can simulate the body shape of different consumers and replace the consumers to try on the clothes remotely. The fitting results will be fed back to the consumers as a reference to improve the efficiency of online clothing purchase or customization. Compared with the virtual fitting, the fitting effect of the fitting robot is closer to the real fitting, and the fitting results are more accurate. Therefore, in this paper, we study the fitting results' extraction and feedback based on the fitting robot.

The accuracy of the fitting results is crucial to the consumers. The fitting results are usually fed back in the form of pictures. The consumers can only get the visual effect through these pictures, and they cannot obtain the actual feeling of clothes-wearing and the comfort level of each part of the clothes. Therefore, to make the remote fitting results more accurate and effective, we design a comfort feedback method for remote fitting to obtain and feed back the comfort information.

This work mainly includes the selection of comfort measuring positions, the design of a pressure measuring device, the preprocessing of measured data, the establishment of a comfort conversion model, and the real-time feedback of comfort information.

2 RELATED WORK

With the rapid development of garment e-commerce, some universities and research institutions have begun to research fitting robots in recent years. For example, Fits.me Company developed a robotic mannequin named Fitbot [1][2][16], Bangkok University designed a male tailoring dummy robot for the tailoring services [14], Kunihiro Ogata and other Japanese researchers designed a dummy humanoid robot to simulate several trunk postures and abdominal shapes [30], Hong Kong Polytechnic University developed an interactive robotic

mannequin for fashion industry [31][32], Zhejiang University developed robotic mannequins by using a belt net [19][35] or layered panels [8][9][10][11]. Among the above studies about fitting robots, only Fitbot's research includes the acquisition of fitting results, and the fitting results are only in the form of pictures. Therefore, it is necessary to study other forms of fitting results.

Clothing comfort is an important factor in determining whether a garment fits. It includes three elements: pressure comfort, tactile comfort, and thermal-wet comfort [23]. Since we mainly want to solve the problem of inappropriate clothing size, this paper considers mostly pressure comfort.

Many researchers have done studies on the pressure comfort of the garment. Wang et al. defined and calculated the comfort threshold and the difference threshold of clothing pressure [34]. Meng and Zhang studied the principle of clothing pressure and the relationship between fitting and clothing comfort [23]. Nakahashi et al. showed that clothing pressure affects skin blood flow, leg temperature, and the heart rate [27]. Some researchers used finite element analysis or other methods to simulate the changes of clothing pressure [24][37], so as to analyze the fit level of clothes and apply it to virtual fitting [20][21] or 3D garment design [18].

At present, there is no research on the measurement and feedback of the garment comfort based on the fitting robot. Therefore, in this paper, we study the comfort feedback method to get and feed back the comfort information based on a fitting robot for remote fitting.

3 COMFORT FEEDBACK METHOD

The comfort feedback method consists of a pressure measuring device, a comfort conversion model, and a feedback interface. Figure 1 shows the application flow of this method. Firstly, the pressure measuring device collects the pressure values from the key positions of the robot. Then, the pre-processed pressure values are transformed into comfort information by using the comfort conversion model. Finally, the comfort information is fed back to consumers through the interface.

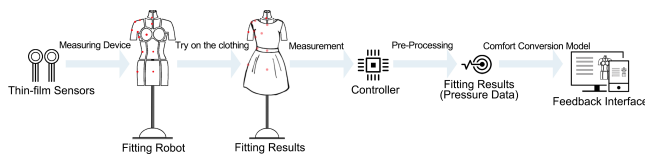


Figure 1: Application Flow of the Comfort Feedback Method.

3.1 Measuring Positions of Garment Pressure

Since different parts of the human body have different clothing comfort requirements, measuring positions of pressures has a significant influence on determining the comfort level of the clothing. So, it is necessary to choose an appropriate installation position for each pressure sensor.

Valid questionnaires from 170 female consumers in the university were collected. In the questionnaire, each participant was asked to select three of the eight body parts that required higher wearing comfort. Figure 2 (a) shows the selection results of the eight body

parts. Moreover, seven parts with a higher proportion were selected for a further selection of sensing positions. For example, the 101 participants who selected the shoulder part in the previous step also chose two more sensitive positions among five positions. Figure 2 (b) shows the selection results of the shoulder part.

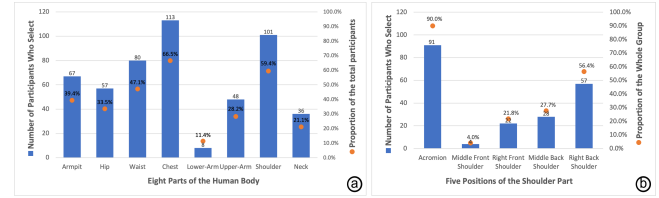


Figure 2: Selection Results of (a) Eight Body Parts and (b) Five Positions on the Shoulder Part.

According to the statistical results of the survey and the symmetry of the human body, 13 positions with higher requirements for wearing comfort were selected on one side of seven parts of the human body, including the front neck, right acromion, right back shoulder, upper arm, armpit, front chest, back, front waist, right waist, back waist, abdomen, hip and right side of the hip. The blue dots in Figure 3 (a), (b) and (c) show the installation positions of 13 sensors on the robot.



Figure 3: Installation Positions of 13 Sensors: (a) 6 Positions, (b) 3 Positions, (c) 4 Positions. (d) Installation of 6 Pressure Sensors on the Front Part of the Robot.

3.2 Design of a Pressure Measuring Device

The pressure measuring device mainly includes a controller, 13 pressure sensors, and 13 analog circuit modules. An Arduino Mega board is used to design the controller with 16 analog interfaces that can simultaneously measure 13 pressure sensors. The Arduino board is equipped with a Bluetooth module connected to the computer side and uploads the measured data in real-time.

To avoid affecting the fitting robot's surface, we adopt the Flexi-force film sensors to measure the garment pressures. The thickness of this kind of sensor is only 0.127mm, and it makes almost no effect after being installed on the surface of the robot. Based on the normal range of garment pressure, we selected the Flexi-force A201 sensor with a range of 0-4400 mN. The resistance of the sensor is greater than 5 MΩ at no load. Moreover, when applying the maximum pressure of 4400 mN, the unit of resistance is also kΩ. Therefore, the amplification circuits are needed to amplify the signals of the sensors. Since 13 sensors are distributed in different

parts of the robot, we use an integrated analog circuit module to avoid complicated circuit wiring. Its length is only 1.7cm, and it can be mounted directly at the end of the pressure sensor. Figure 3 (d) shows the test and installation of 6 pressure sensors on the front part of the robot.

Finally, 13 pressure sensors are installed in the preselected positions on the surface of the robot. Each analog circuit module at the end of the sensor is installed inside the corresponding panel. Arduino Mega transmits the measured data to the computer in real-time through the Bluetooth module for further processing and feedback. When the garment pressure exceeds the measurable range, the controller will send an instruction to stop the deformation of the robot to prevent the robot structure and the sensors from being damaged.

3.3 Pre-Processing of Measured Data

The data obtained by the pressure measuring device needs to be pre-processed and then converted. The pre-processing of the measured data includes filtering the original data, converting the voltage signals to the pressure values, and eliminating the invalid data sets.

Each sensor's measured data is filtered to remove the random errors and prevent jittering when drawing in the interface. At present, there are mean-value filtering method, median filtering method, adjacent resampling method, and low-pass filtering method based on the frequency domain, and so on. In this research, we use the mean-value filtering method to process the original data.

According to the measuring circuit principle of the Flexiforce sensor, the pressure is linearly related to the output voltage of the amplifying circuit, so the measured pressure can be calculated.

After filtering the original data and converting the pressure values, the invalid data sets need to be eliminated. Arduino will read the signals from 13 sensors simultaneously, and these 13 parameters are formed into a data set, which will be sent to the Processing for the feedback. Due to signal interference or random error, a process is added to examine and eliminate incomplete data sets.

3.4 Establishment of Comfort Conversion Model

To obtain the comfort level, we need to establish the relationship between the pressure values and comfort levels. Since different users have different feelings about the same garment pressure, it is difficult to match a pressure value with a certain comfort level, so the comfort model is established by matching a range of pressure values with a comfort level.

We collected the comfort feelings of 27 female subjects under different pressures, and the results show that 72.4% of the pressure ranges the subjects felt comfortable with are between 140 mN and 270 mN, and we preliminarily set this range as the comfort level *Comfortable*. The pressure sensor's effective measuring area is $6.94 \times 10^{-5} \text{ m}^2$, so the corresponding intensity of pressure is 2.02 kPa - 3.89 kPa. Wang et al. measured 9 body parts of 10 subjects, and the results show that most of the pressure comfort thresholds of these parts are between 2000 Pa and 3500 Pa [34], which is relatively close to our results. Moreover, 63.8% of the pressure ranges the subjects felt restrained are between 400 mN and 680 mN, and the corresponding intensity of pressure is 5.76 kPa - 9.80 kPa, so we preliminarily set

this range as the comfort level *Strained*. Only one measurement part of one subject has a pressure comfort threshold greater than 5000 Pa in the result of [34], which is similar to our results. When the pressure increased to 680 mN, 70.4% of the subjects already felt tight, and the corresponding intensity of pressure is 9.80 kPa, so we preliminarily set this pressure as the start of the comfort level *Tight*. When the measured pressure is 800 mN, the corresponding intensity of pressure is 11.53 kPa, and it looks the clothes are too tight, so we choose it as the warning value.

Consumers have no specific concept of pressure values, and visual feedback is very weak when comfort information is displayed only by the comfort level. Color is an expressive element that can affect people's intuitive feelings [22][28]. For example, red makes people feel tension and strength, and green makes people feel peace and stability. We use different colors to represent the pressures and use the gradient of colors to represent the pressures' change. Since a comfort level represents a pressure range, the color gradient can intuitively show the pressures change at the same comfort level. According to the current study of color emotion [7] and daily life experience, we define green as the starting color of the comfort level *Loose*, yellow as the starting color of the comfort level *Strained*, and red as the starting color of the comfort level *Too Tight*. If the pressure changes from 0 to 800 mN, the corresponding color changes from green to yellow and then to red. When the pressure changes from 0 to 400 mN, the RGB values of the corresponding color can be calculated according to Equation (1). F_{Flexi} is the pressure applied to the sensor, and the unit is mN.

$$0 \leq F_{Flexi} \leq 400 : \begin{cases} R = \frac{F_{Flexi}}{400} * 255 \\ G = 255 \\ B = 0 \end{cases} \quad (1)$$

Similarly, when the pressure changes from 400mN to 800mN, the RGB value of the corresponding color can be calculated according to Equation (2).

$$400 \leq F_{Flexi} \leq 800 : \begin{cases} R = 255 \\ G = 255 - \frac{F_{Flexi} - 400}{400} * 255 \\ B = 0 \end{cases} \quad (2)$$

Based on the relationship between the pressure ranges and the comfort levels and the mapping relationship between the pressure values and the colors, we established a preliminary comfort conversion model for the measuring device. Figure 4 shows the content of the comfort conversion model. The model is used to convert one pressure value into a comfort level and a color. It is an essential basis for the conversion and feedback of comfort information.

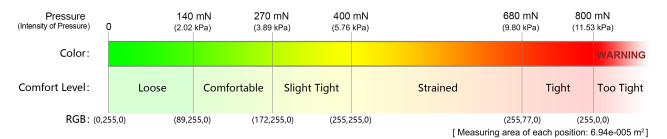


Figure 4: Comfort Conversion Model for the Designed Pressure Measuring Device.

3.5 Feedback of Comfort Information

The comfort information transformed by the conversion model will be fed back to the interactive interface. The interface is preliminarily built by Processing software. It includes a display module of pressure values and comfort levels, a color display module of measurement positions, and a display module that shows the change process of pressures. Figure 5 shows the interface's warning status when the comfort level of at least one position is Too Tight.



Figure 5: Feedback Interface of Comfort Information

Through the real-time feedback of pressure values and comfort levels, consumers are provided with more accurate fitting results. Through the color feedback and the dynamic changing process of pressures, consumers can get an intuitive comfort feeling and a dynamic interactive experience.

4 RESULTS

Based on the pressure measuring device, the comfort conversion model, and the feedback interface, we establish a preliminary comfort feedback method to acquire and feed back the comfort information of remote fitting.

After customizing two dresses for consumers, the designer used the fitting robot to try on each customized dress to ensure the customization effect, and at the same time, the comfort feedback method was used to obtain and feed back the comfort information. Figure 6 shows the comfort result. Figure 7 shows the combination of the comfort information fed back from the system and the fitting pictures, which demonstrates that each position's comfort results are intuitive.

We collected subjective feelings from consumers after they tried their customized dresses. Table 1 shows the comparison results of subjective feelings and comfort information fed back by the measuring system. The results show that consumer A's subjective feelings in 10 positions are the same as the comfort levels of system feedback, which is 11 positions for consumer B. Furthermore, the overall feeling of both of the customized clothing is comfortable. The comparison results show that the comfort feedback method is effective, but everyone has a large difference in comfort feeling, so more users and clothes need to be tested in the future to verify the method's stability further.

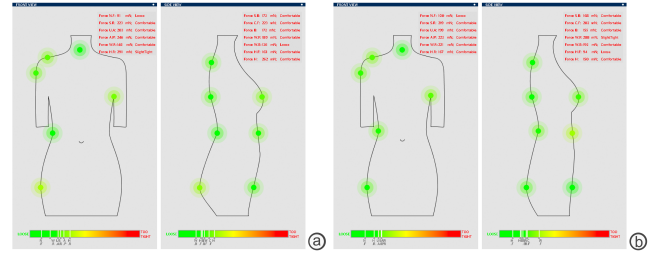


Figure 6: Comfort Information of Two Customized Dresses: (a) Customization A and (b) Customization B.



Figure 7: Customized Dresses and Comfort Information for (a) Customization A and (b) Customization B.

5 LIMITATIONS AND FUTURE WORK

At present, the comfort conversion model was built without subdividing different parts of the human body, but different parts have different comfort perceptions, so we need to further refine the model in future work. Moreover, the amount of data collected from subjects for establishing the comfort conversion model is not large enough. In addition to the clothing pressure, the wearing comfort also is significantly affected by the wearer's physical and psychological feelings, so it is necessary to establish an extensive statistical database of subjective feelings and clothing pressures.

Due to the current pandemic, prototype production, user testing, and data acquisition have been affected. In the future, to ensure the effectiveness and stability of the comfort conversion model and the feedback method, more users and clothes of different styles and materials need to be tested in the verification test.

The fitting robot's surface material is foamed silica gel, which is similar to the surface of the human body, but still has certain differences. Since the comfort conversion model is established based on the clothing pressures on the surface of the human body and used to transform the clothing pressures measured on the robot's surface, the comfort conversion model needs to be further corrected according to the difference between a robot and a human body.

Furthermore, the measuring tests were conducted only with the subjects in a standing posture, but people have various postures in daily life. In future work, it can be considered to measure garment pressures and subjective feelings in sitting and dynamic states.

In the next work, we will also consider adding more measuring positions to make the overall garment's comfort result more

Table 1: Comparison of Subjective Feelings and Comfort Information Obtained by the Measuring System

Positions	Consumer A			Consumer B		
	F (mN)	Comfort Level	Subjective Feeling	F (mN)	Comfort Level	Subjective Feeling
Front Neck	81	Loose	Loose	100	Loose	Loose
Right Acromion	223	Comfortable	Comfortable	219	Comfortable	Comfortable
Right Back Shoulder	172	Comfortable	Comfortable	168	Comfortable	Comfortable
Upper Arm	203	Comfortable	Comfortable	199	Comfortable	Comfortable
Armpit	246	Comfortable	Slight Tight	222	Comfortable	Comfortable
Front Chest	223	Comfortable	Comfortable	202	Comfortable	Comfortable
Back	172	Comfortable	Comfortable	155	Comfortable	Comfortable
Front Waist	189	Comfortable	Comfortable	288	Slight Tight	Slight Tight
Right Waist	145	Comfortable	Comfortable	221	Comfortable	Slight Tight
Back Waist	126	Loose	Comfortable	192	Comfortable	Comfortable
Abdomen	163	Comfortable	Comfortable	94	Loose	Comfortable
Hip	262	Comfortable	Slight Tight	150	Comfortable	Comfortable
Right Hip	291	Slight Tight	Slight Tight	167	Comfortable	Comfortable
Overall	-	-	Comfortable	-	-	Comfortable

objective and effective. Furthermore, when the measured positions increase by a certain number, other unmeasured positions can be simulated, and an intuitive heat map of the whole body's comfort information can be generated for feedback.

6 CONCLUSION

We present a preliminary comfort feedback method in this paper. It is used to obtain and feed back the comfort information of remote fitting based on the fitting robot, thereby improving the effectiveness of remote fitting and reducing the return rate of garment e-commerce. A pressure measuring device is designed to collect clothing pressures from 13 key positions. A comfort conversion model is established to convert pressures into intuitive comfort levels and colors. The interface will feed back the measured pressure values, comfort levels, color display of pressures, and dynamic change process of pressures in real-time. In this way, it provides users with accurate fitting results and a dynamic interactive experience, which is conducive to the development of garment e-commerce, especially in this pandemic period. The preliminary research on this method also demonstrates some research opportunities on garment comfort of the remote fitting.

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